CS 301 · MEETING 12 · A DEEPER LOOK, 2

What Computing Costs

A deeper look: the seconds of this meeting in joules and in dollars, and a model trained at scale.

Three costs of one computation

$$\text{time} = \frac{\text{operations}}{\text{operations per second}} \qquad \text{energy} = \text{power} \times \text{time}$$

$$\text{money} = \text{price per hour} \times \text{hours}$$

- Power in **watts**: joules each second
- 1 kilowatt-hour = 3.6 million joules
- The first cost is the count and the rate of this meeting; the other two follow from it

Watts

processor	power	source
Colab's T4	70 W at most	NVIDIA's data sheet
the CPU of the machine of these slides	55 W base, 157 W at most	Intel
a V100 training GPT-3	330 W, with its share of the server	measured; Patterson et al., 2021

- A rated power is an upper bound: a processor may draw less
- Colab does not publish its CPU's model or power

The joules of one product

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X @ W.T at size 4,096: 137 billion operations
processor                              seconds  watts   joules   operations a joule
CPU of this machine, 32 logical cores    0.288    157     45.2         3.04 billion
Colab's T4: your posts                   0.049     70      3.4        40.3 billion
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- energy = watts × seconds; each processor at its rated power
- At rated power, the T4 has the lower estimated energy

Dollars an hour

	price	source
a T4, rented by the hour	$0.35	Google Cloud, on demand
a V100, rented by the hour	$2.48	Google Cloud, on demand
Colab's T4	free, when one is spare	Google Colab
electricity, a kilowatt-hour	15 cents	U.S. average, July 2026 (EIA)

- A T4-hour of electricity: 70 W for an hour, 0.07 kWh, about one cent
- The rent is about 30 times the electricity

The dollars of one product

on a rented T4, at $0.35 an hour, in cents	rent	electricity
one hour	35	1
one product at size 4,096	0.00047	0.000014
a trillion operations, at your rate	0.0034	0.0001
the layer of the notebook: 94 billion	0.00032	0.0000097

- A trillion operations on a T4: about three thousandths of a cent
- Cheap per operation; the counts of real models are enormous

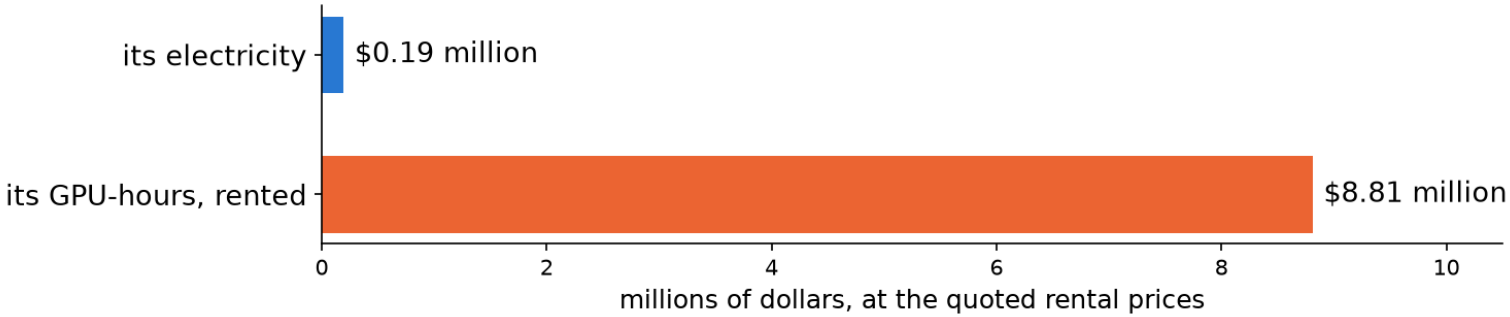
A model trained at scale: GPT-3

operations to train it	3.14×10^{23}
processors	10,000 V100 GPUs
time	14.8 days
power of each, with its share of the server	330 W
energy	1,287 MWh
carbon dioxide, and its equivalents	552 tonnes

- Patterson et al., *Carbon Emissions and Large Neural Network Training*, 2021

GPT-3 in the units of this meeting

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GPT-3: 3.14 x 10^23 operations
on one T4 at your rate:                3,526 years
on 10,000 V100s, as it was trained:   14.8 days, 3.55 million GPU-hours
its electricity, 1,287 MWh, at today's U.S. average price:  $192,921
its GPU-hours at the quoted list price for a V100:          $8,808,960
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- On your T4: thousands of years
- Most of the bill is the rent of the processors, not the electricity

What decides the cost

$$\text{cost} \approx \text{count} \times \text{seconds for one operation} \times \text{price of a second}$$

- The count: O( ), the same on every machine
- The seconds for one operation: the machine, the order, the memory
- The price of a second: watts, and dollars an hour
- A faster processor can cost more an hour and less per operation